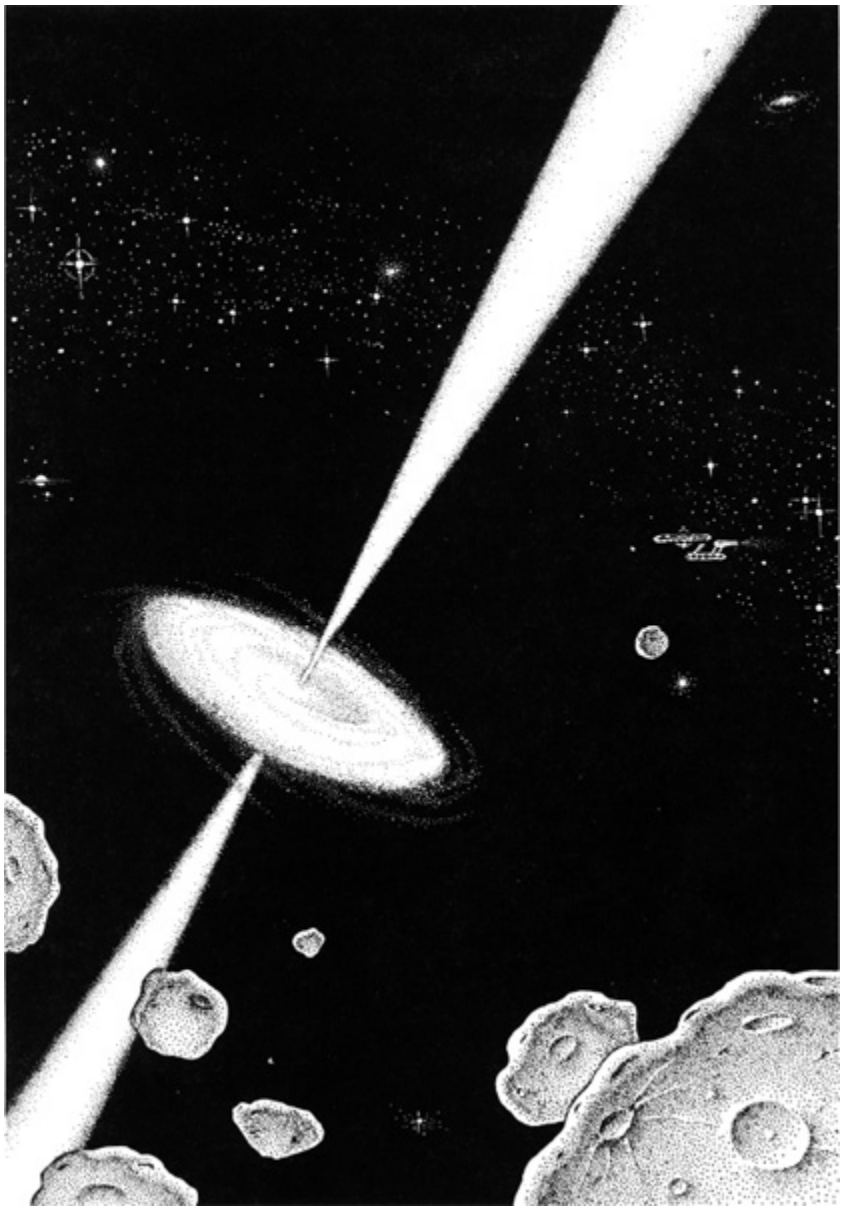


P.5 The quasar 3C273: a 2-billion-solar-mass black hole encircled by a doughnut of gas ("accretion disk") and with two gigantic jets shooting out along the hole's spin axis.

You do not want to arrive at some chosen hole and discover that other beings have already built a civilization around it; so instead of aiming your starship at a rapidly spinning hole that already exists, you aim at a star system which will give birth to a rapidly spinning hole shortly after your ship arrives.

In the Milky Way's Orion nebula, at the time you left Earth, there was a *binary star system* composed of two 30-solar-mass stars orbiting each other. DAWN has calculated that each of those stars should have imploded, while